

Bazy danych i systemy DBMS: wprowadzenie Introduction to databases and Database management systems

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Database course contents: part I

- Introduction
- How structured data are managed:
 - Relational databases (*pol. relacyjne bazy danych*)
 - Database design and normalisation (*projektowanie i normalizacja baz danych*)
 - Entity-relationship modelling
 - SQL language (język SQL)
 - Performance aspects and indexes
- How to keep our data secure: data security, integrity,...
- How to let multiple users modify data at the same time: transactional processing (*przetwarzanie transakcyjne*)
- How to develop the code creating and modifying data:
 - On the server side: stored procedures (*procedury składowane*), triggers (*wyzwalacze*), ...
 - On the client side: Java-based and .NET-based client applications and database APIs

Database course contents: part II

Overview of other aspects of databases and database systems, including introduction to:

- Using data for analytical needs: data warehouses (*hurtownie danych*), OLAP systems and Business Intelligence
- Managing maps and geodata: GIS systems (*systemy informacji przestrzennej*)
- Being a database administrator: Oracle database as an example of modern Database Management System
 - architecture
 - security
 - administration aspects
- Managing large volumes of (possibly unstructured) data: Big data storage and Apache Hadoop
- Managing multimedia data and linking it with AI: vector databases
- Managing large volumes of semi-structured data: NoSQL platforms
- ...

Resources

- Lecture notes and lab scripts
- Books:
 - S. Hoberman, Data Modeling Made Simple, 2nd Ed., 2016, Technics Publications
 - S. Sumathi, S. Esakkirajan, **Fundamentals of Relational Database Management Systems**, Springer Berlin Heidelberg, 2007
 - J. Viescas, M. J. Hernandez, *SQL Queries for Mere Mortals: A Hands-On Guide to Data Manipulation in SQL*, Addison-Wesley Professional, 2014, (Zapytania w SQL: przyjazny przewodnik, Helion, 2015)
 - Louis Davidson, Jessica M. Moss, **Pro SQL Server 2012 Relational Database Design and Implementation**, Apress, 2012
 - R. Elmasri, S. B. Navathe, *Fundamentals of Database Systems*, Addison-Wesley, 2004 (also in Polish: Wprowadzenie do systemów baz danych, 2019 by Helion)
 - R. Kimball, M. Ross, *The Data Warehouse Toolkit*, Wiley, 3rd Ed., 2013
 - P. Atkinson, R. Vieira, **Beginning Microsoft SQL Server 2012 Programming**, John Wiley & Sons, 2012
 - D. Bales, M. Rosenblum, **Beginning Oracle PL/SQL**, Apress, 2015

Podkreślenie – dostępne w Bibliotece Głównej/Underlined – available in Main Library
Pogrubienie – ebook w Bibliotece Głównej/**Bold** – ebook available via Main Library

Selected databases E-books available via Main Library

- P. Atkinson, R. Vieira, Beginning Microsoft SQL Server 2012 Programming, John Wiley & Sons, 2012
- D. Bales, M. Rosenblum, Beginning Oracle PL/SQL, Apress, 2015
- S. Sumathi, S. Esakkirajan, Fundamentals of Relational Database Management Systems, Springer Berlin Heidelberg, 2007
- Louis Davidson, Jessica M. Moss, Pro SQL Server 2012 Relational Database Design and Implementation, Apress, 2012
- Malcher, M., Kuhn, D., Pro Oracle Database 18c Administration: Manage and Safeguard Your Organization's Data, Apress, 2019

These and other books are available at <http://www.bg.pw.edu.pl>

You may also use this address: [http://sfx-](http://sfx-48tuw.hosted.exlibrisgroup.com/48tuw/ebooksearch?lang=pol)

[48tuw.hosted.exlibrisgroup.com/48tuw/ebooksearch?lang=pol](http://sfx-48tuw.hosted.exlibrisgroup.com/48tuw/ebooksearch?lang=pol). In spite of the fact that most recent editions are typically not included in ebbok collection of Main Library, these books are more than helpful to learn the basics of SQL, database design, E-R modelling and related aspects.

Learning databases in AI era

- Are recent developments in AI and tools such as ChatGPT and GitHub Copilot helpful for Data Science/IT professionals?
- Yes, we can generate a lot of code (and not only code)...
- Does it mean Data Science/IT professionals working in business:
 - have more free time?
 - have as many job opportunities as before?
- Companies report savings thanks to AI in software development:
 - <https://developers.slashdot.org/story/24/08/25/0049230/amazon-ceo-ai-assisted-code-transformation-saved-us-4500-years-of-developer-work>
- What do savings mean for companies?
- Is AI an ally or a competitor of IT/Data Science professionals?
- Can IT professionals be less prepared, or should they be better prepared than before to be still needed?
- Demand for junior IT professionals has largely reduced.
- To be needed, Data Science/IT professionals have to be even more professional than before. One of the major expectations is to **carefully** evaluate and **improve** solutions suggested by AI.
- Does it require in-depth knowledge and skills?

Developing database systems is something that will require major human contribution. Why? Because we need to capture requirements and match the expectations of business professionals. Developing and using databases means combining management and technical aspects.

Learning databases during Databases in Data Analytics

- In the AI era, to be needed, Data Science/IT professionals must know even more before.
- Hence,
 - As trivial tasks are automated with AI, it is even more important to:
 - Know and be able to apply complex solutions
 - To understand how to adapt them to the unique needs of enterprises
 - Let us use also our lecture classes to match this objective:
 - Lecture $\neq$ slides. We will discuss more than is on the slides
 - Lecture $\neq$ prerecorded presentation - questions not only can be asked, but also should be
 - Our time during the labs is limited. Limiting the scope of the course and learning basic solutions only would not be ... fair
 - **Hence, let us use the lectures to learn more than can be done during the labs and gain the “big picture”**
 - **As there is much to discuss, if anything sounds unclear, please do not hesitate to ask questions.**

To help you with learning databases, we will arrange anonymous online quizzes (at the end of lecture meetings starting from the second one) and more...

Bazy danych: zasady zaliczeń

- Laboratorium:
 - maks. 70 punktów
 - punkty za testy wstępne z materiału wykładów (na dwóch laboratoriach)
 - punkty za zadania punktowane Z1, Z2A, Z2B, Z3
 - punkty za aktywny udział w zajęciach
 - obecność na 12 zajęciach obowiązkowa (konieczna do zaliczenia)
- Egzamin z całości materiału (w tym całości materiału wykładu)
 - maks. 30 punktów
- Do zaliczenia wymagane co najmniej 50% z każdego z zadań, czyli:
 - co najmniej 50% punktów z każdego z zadań laboratoryjnych Z1, Z2A, Z2B, Z3
 - co najmniej 15 punktów z egzaminu
- Ocena końcowa zależy od łącznej liczby punktów:
 - $[0,50) = > 2,0$
 - $[50,61) = > 3,0$
 - $[61,71) = > 3,5$
 - $[71,81) = > 4,0$
 - $[81,91) = > 4,5$
 - $[91,100] = > 5,0$

- W przypadku uzyskania z testu wstępnego co najmniej 80% punktów, do sumarycznej liczby punktów uzyskanych z przedmiotu doliczane są 3 punkty z tytułu jednego testu.
- W przypadku uzyskania poniżej 60% punktów z testu, sumaryczna liczba punktów z przedmiotu jest redukowana o 5 punktów z tytułu jednego niezaliczonego testu.

Punktowane zadania laboratoryjne

Zadanie	Termin	Maks. liczba punktów
Z1. Projektowanie baz danych, zapytania SQL SELECT	Laboratorium 7	25
• Z2A. Kod dostępu do bazy danych, INSERT/UPDATE/DELETE (10 punktów) • Z2B. Weryfikacja jakości proponowanego rozwiązania (klient bazy danych/SQL) (10 punktów)	Laboratorium 11	20
Z3. SQL DDL, indeksy, transakcje, procedury składowane	Laboratorium 14	20
Aktywność na zajęciach - rozwiązywanie i prezentowanie wyników krótkich zadań oraz dod. scenariuszy.	Lab. 1-13 z wyjątkiem zadań punktowanych	5

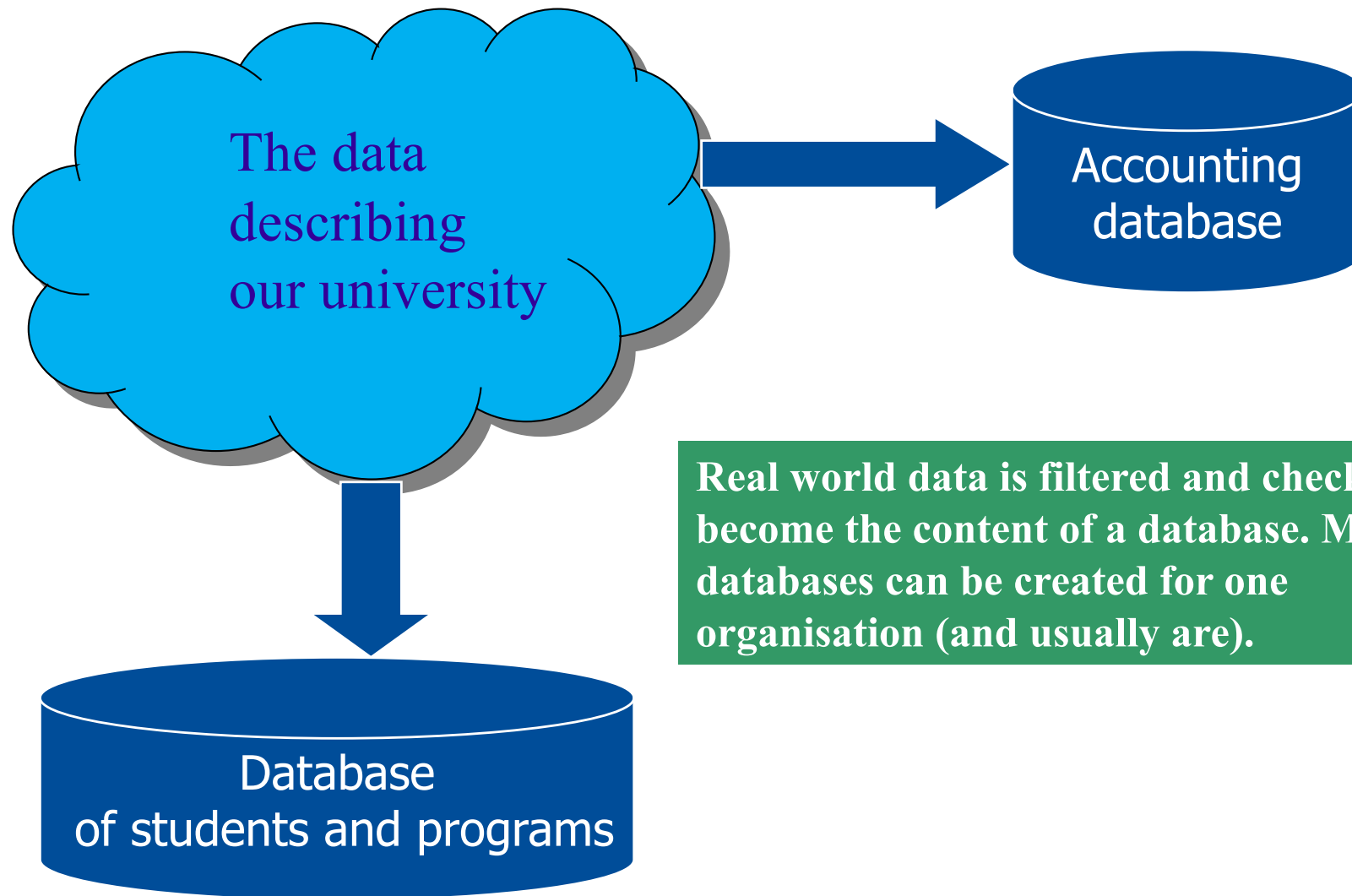
Możliwe jest przesunięcie terminu w ramach semestru o 1-2 tygodnie, w porozumieniu z prowadzącym zajęcia danej grupy laboratoryjnej.

Finalna liczba punktów za dane zadanie może zależeć od dodatkowej odpowiedzi ustnej. Jeśli wystąpi taka potrzeba, organizowane jest 1 laboratorium poprawkowe. Uwaga: slajdy zawierają kluczowe informacje. Szczegółowe zasady zaliczenia są dostępne w regulaminie przedmiotu w Usos. Ewentualne pytania/uwagi do regulaminu proszę zgłaszać drogą mailową najpóźniej dwa dni przed drugim wykładem.

Database (Baza danych)

- Definition:
 - *set of positive facts about real world from the universe of discourse stored in persistent manner.*
 - *~Zbiór pozytywnych faktów na temat świata rzeczywistego z pewnego zakresu tematycznego przechowywanych w sposób trwały.*
- Typical representation:
 - Set of disk files containing the data
 - The files are organised so as to speed up information retrieval and update
 - Relational database i.e. a database in which data is organised in tables
 - Typically the data of multiple tables is placed in one disk file

Database – an example



Real world data is filtered and checked to become the content of a database. Many databases can be created for one organisation (and usually are).

Intuition: database of buildings

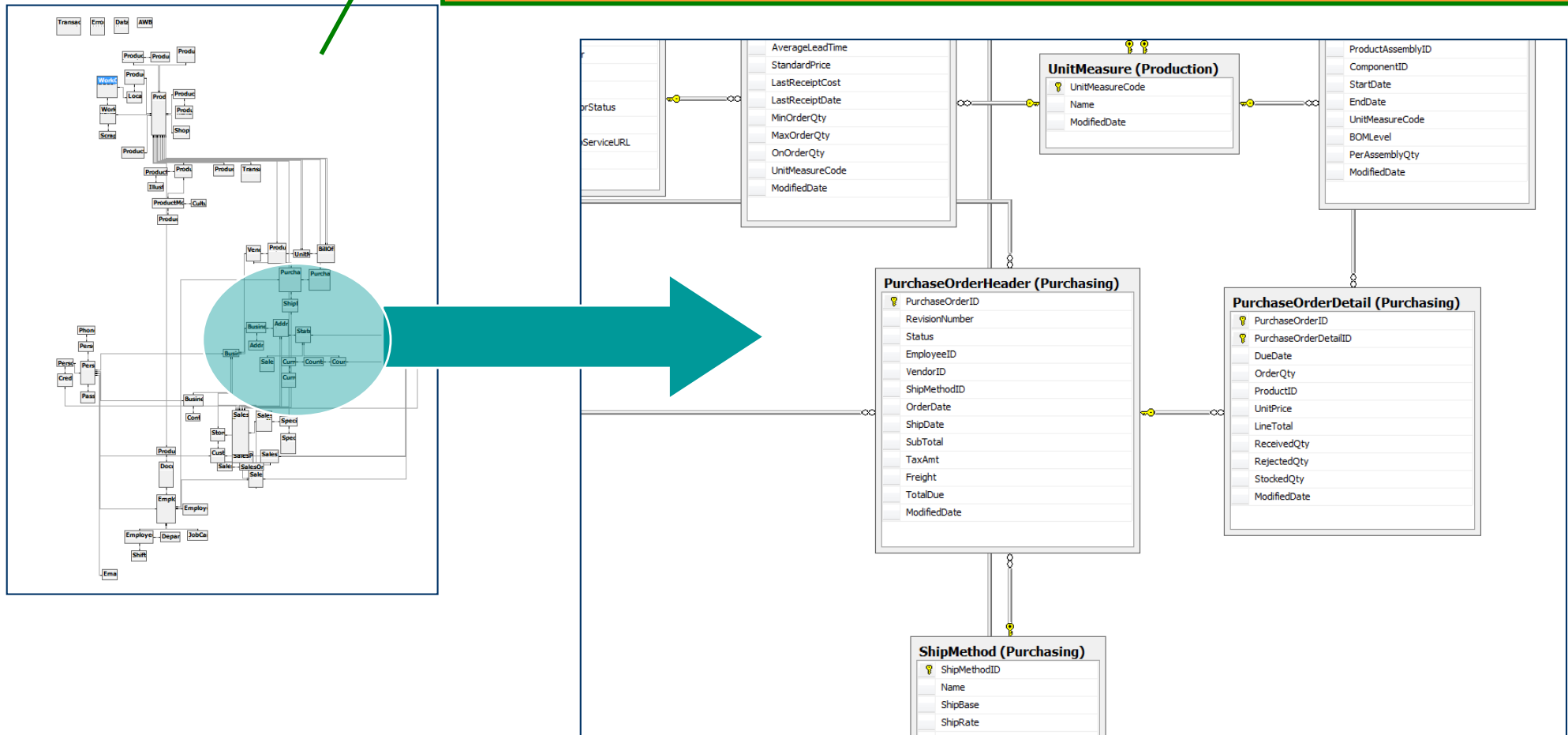
Building name	City	Street	...	
Main Building	Warszawa	Pl.		
Chemistry	Building	Room	Area	...
	Main Building	213	45	
Physics	Main Building	215	212	
	. . .	. . .		
	Chemistry	452	350	

For most databases i.e. *relational databases* the content of a database can be imagined as a set of tables containing the description of real world facts. Each table contains many rows called *records*.

Positive facts are stored e.g. only existing rooms are listed for each building

Intuition: database

Typically tens or hundreds of tables comprise on a single database. These are linked by relations e.g. room record refers to the building record in which the room is located

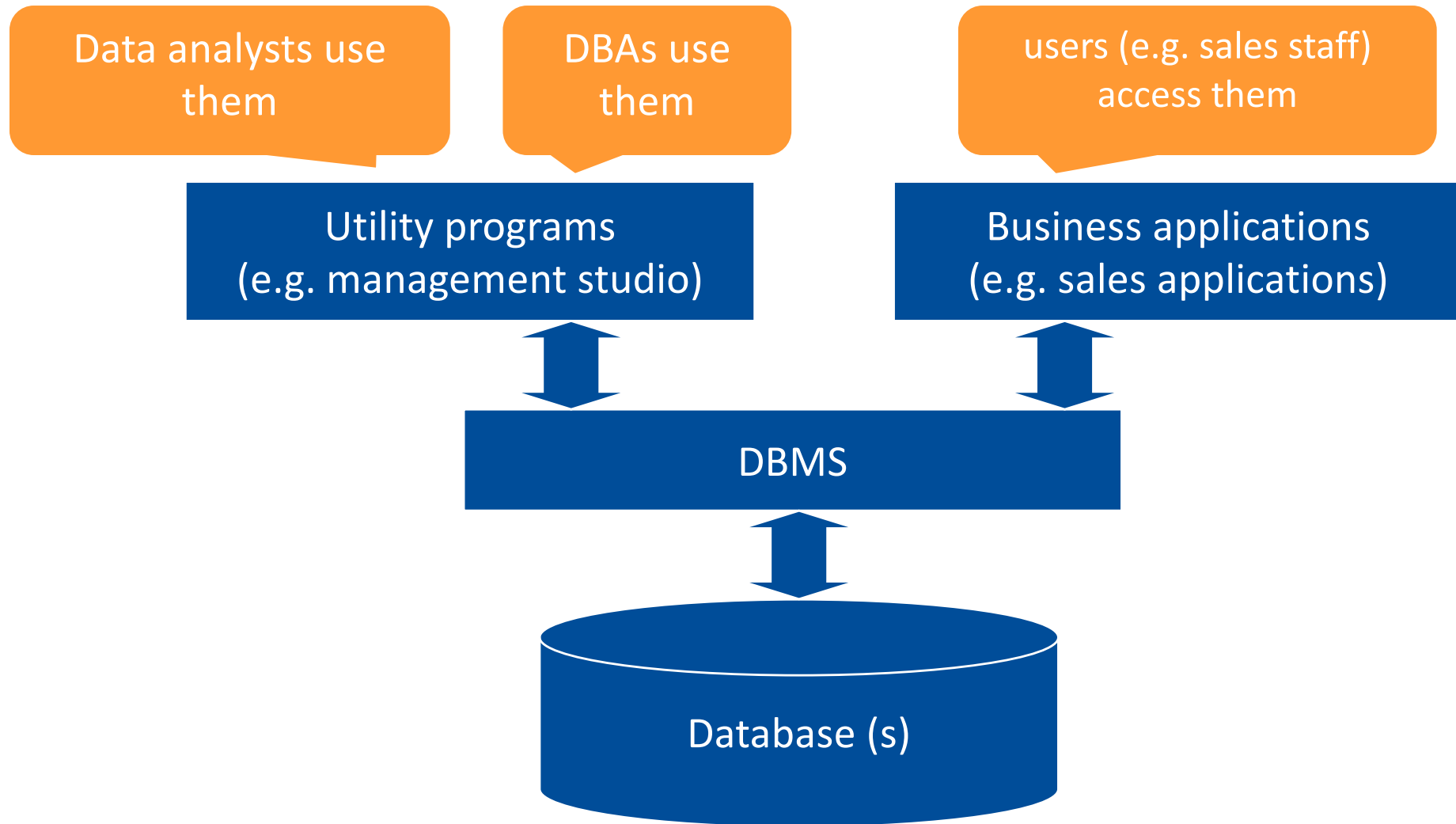


Database management system

- Database management system (DBMS) (*System Zarządzania Bazą Danych*) – software system or library that:
 1. Interacts between user applications and database to provide the access to the data by:
 - Saving data into data files
 - Retrieving data from data files by answering possibly complex SQL queries
 2. Supports administrative tasks of database administrator (DBA, *administrator baz danych*), which includes creating and manipulating databases
 3. Provides add-on functionalities such as online backups, permission system, replication, indexing etc.

While all DBMS platforms provide data storage and retrieval, commercial DBMS platforms offer extensive state-of-the-art security, performance and high availability.

Database and DBMS: a typical layout of client-server architecture



Setting up your own DBMS instance

- Your own DBMS instance will be crucial for doing laboratory tasks and preparing your home works
- In a separate presentation (available in the files of your Databases team in Ms Teams) you will find a tutorial **db_lab_1b_environment.pdf** on:
 - how to get, install and configure Ms SQL Server:
 - On Windows
 - On Linux (in case you strongly prefer Linux over Windows)
 - What to do in case your environment is not strong enough to use recent version of Ms SQL Server
 - how to get and set up Ms Visual Studio, which will be also needed to follow one of the scenarios
- Please note that due to limited hardware resources, unfortunately online access to DBMS instances provided by the faculty will be limited and not always possible
- Hence, having your own environment is strongly recommended

Should you have any problems with setting up your Ms SQL Server and Ms Visual Studio for Databases labs, please report them during the first two labs.

Summary

- Databases serve the need for storing data in an efficient manner making queries and data modifications possible
- The most common category of databases are relational databases
- These are recently accompanied by NoSQL platforms such as MongoDB. The growth of these platforms is largely due to the need to process unstructured and semi-structured data of possibly large size (Big data)
- Still, NoSQL/Big Data platforms are not a replacement for relational database management systems
- The most advanced DBMSs are commercial relational DBMS platforms

Pytania ?

Zapraszam w trakcie wykładu
i w trakcie konsultacji

Miejsce i termin konsultacji
- na stronie:

[https://pages.mini.pw.edu.pl/
~grzendam/pl/dydaktyka.html](https://pages.mini.pw.edu.pl/~grzendam/pl/dydaktyka.html)



Questions?

I am looking forward to your questions during the lectures (including now) and during my office hours

For details on my office hours, please check my website:

https://pages.mini.pw.edu.pl/~grzendam/eng/dydaktyka_eng.html





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